An Efficient Measurement Method for Fabric Defect Size Based on Multilight-Source Computer Vision

Shangyan Zhang, Jian Zhou, Jun Xiang, and Ruru Pan

***Abstract*—Defects are inevitable on fabric surfaces during the weaving process, and their size is the primary criterion for fabric quality rating. Existing methods primarily rely on computer vision under a single lighting condition to isolate defect pixels from the background, enabling defect contour localization for size measurement. However, due to limited illumination, the shallow features of certain defects remain weak and difficult to detect, even for human observers, thereby compromising the accuracy of the measurement. Moreover, adaptability to the complexity and diversity of fabric defects is constrained by reliance on a single method. To address these challenges, a multialgorithm collaborative measurement system is proposed, which is supported by an image acquisition platform equipped with multiple illumination sources and designed to decouple and adapt to complex defect measurement tasks. The system, designed for production line deployment, acquires images via alternating flashes from multiple light sources, simulating human vision and enhancing defect saliency to provide high-quality inputs for parallel localization algorithms. Tailored algorithms corresponding to each illumination source consist of training and testing phases for deployment. During training, a normal hyperplane is constructed from fabric images under diffuse illumination, serving as the baseline for defect localization, while the other illuminations assist in detecting defects that diffuse illumination cannot capture, enabling complementary measurement. To evaluate the accuracy of size measurement, the proposed E_ϵ is defined as the ratio of the absolute error between the predicted and actual defect dimensions in the warp or weft directions to their sum. Experimental results demonstrate that the system achieves a size measurement error of 0.16, which is lower than that of existing methods, validating its effectiveness. Additionally, the system is adaptable to fabrics of various specifications and can calculate defect sizes in real time on the production line for practical applications.**

***Index Terms*—Computer vision, fabric defect size, image processing, measurement system, multilight-source illumination.**

I. Introduction

In RECENT years, intelligent automation technology has developed rapidly and achieved significant research results in manufacturing companies, especially in product defect size measurement on production lines [1]. However, textile companies still rely on inefficient manual vision to measure the size of defects on the textile surface as an important evaluation indicator for textile quality grading before selling the fabrics, which indicates the urgent need for intelligence in textile manufacturing companies [2].

During the production process of finished fabrics, occasional mechanical failures, yarn breakages, or uneven dyeing may occur, resulting in defects on the fabric surface, such as knots and fuzz balls. These defects are a key factor in reducing fabric quality, and defects of different sizes lead to varying degrees of quality degradation, ultimately determining the fabric's potential applications. During the quality inspection phase preceding the industrial application of fabrics, inspectors are required to sustain prolonged concentration and rely on visual observation of dynamically conveyed surfaces to promptly determine whether defects are present within their field of view. Once a defect is detected, production must be halted for dimensional measurement. Accordingly, fabric quality assessment primarily involves two processes: defect localization and dimensional measurement.

The primary challenges in fabric defect size measurement involve the targeted localization of defect information and the development of real-time contour detection algorithms that enable accurate size estimation while emulating human visual perception. Due to the inherently soft texture of most fabrics, surface defects exhibit diverse morphologies, which not only increase the difficulty of defect localization for the human visual system but also pose challenges for image acquisition systems based on cameras. Under single-source illumination, certain defect features provide only weak visual stimuli, resulting in incomplete semantic perception by the human visual system and increasing the likelihood of missed detections. Such omissions not only lead to inaccurate measurements and biased quality evaluations but also cause significant economic losses for enterprises. Although adjusting illumination conditions can enhance defect visibility, frequent changes hinder human adaptability. Moreover, minor surface deformations are often misinterpreted as defects, while the downtime required for verification further reduces production efficiency. To overcome these limitations, computer vision-based approaches have been explored [3], [4]. As illustrated in Fig. 1, defect contours must first be determined and defect sizes subsequently calculated to reflect fabric quality, excluding defects that are missed due to single-source illumination. Consequently, adopting a multisource illumination-based approach for fabric quality assessment is essential, as it minimizes missed detections, enables automatic illumination switching, and supports algorithm-driven defect localization, dimensional measurement, and quality evaluation without interrupting production.

Received 4 July 2025; revised 25 September 2025; accepted 11 October 2025. Date of publication 17 November 2025; date of current version 21 November 2025. The Associate Editor coordinating the review process was Dr. Xiangchen Qian. *(Corresponding authors: Ruru Pan; Jun Xiang.)*

The authors are with the College of Textile Science and Technology, Jiangnan University, Wuxi, Jiangsu 214000, China (e-mail: Zhang_sy358@163.com; jzhou@jiangnan.edu.cn; skyjun12@163.com; prrsw@163.com).

Digital Object Identifier 10.1109/TIM.2025.3628419

1557-9662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

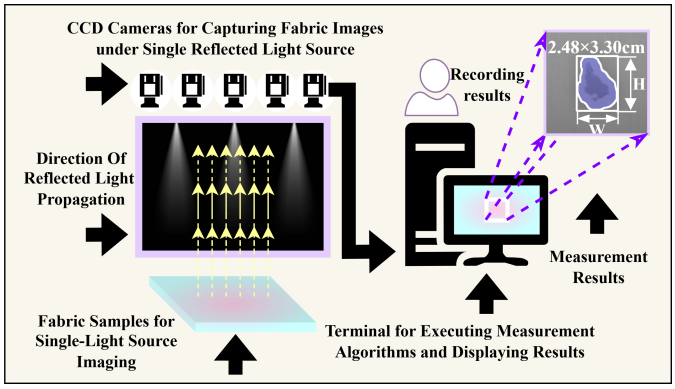


Fig. 1. Fabric defect size measurement system relies on single-light illumination. In fabric manufacturing, defect size, represented by the W (width) and H (height) of the minimum bounding rectangle, serves as a key quality indicator. Measurements based on single-light illumination often face accuracy issues, with missed and false detections leading to incorrect measurements and quality evaluations, resulting in economic losses. The measurement results are calculated based on the pixel dimensions corresponding to the real-world size, expressed in centimeters.

In recent years, deep learning techniques have been extensively studied and are primarily categorized into supervised, self-supervised, and unsupervised learning approaches. Considerable efforts have been devoted to improving the effectiveness of neural networks for object localization in industrial production [5], [6], [7], [8]. Supervised learning-based methods rely on large-scale prior knowledge bases to regress each target as accurately as possible and provide feedback in the form of object categories and their corresponding location coordinates [9]. Furthermore, to fully exploit the potential of unlabeled data and reduce the high costs associated with labeled datasets, pretraining features based on deep networks and self-supervised learning have been widely investigated and successfully applied in industrial settings [10], [11]. For industrial fabric quality inspection, supervised learning approaches demonstrate robustness for most defects; however, their effectiveness is limited by the high cost of annotation, the scarcity of defect samples, and the difficulty of detecting unknown defects, all of which collectively restrict their performance. In addition, unsupervised methods do not rely on prelabeled data; however, inference speed remains a critical challenge, particularly under multi illumination conditions where the execution efficiency of algorithms under each light source constitutes a major factor for their practical deployment in industrial settings. Therefore, major challenges in defect size measurement for quality inspection involve real-time processing of dynamically conveyed images for high-speed production and precise separation of defect pixels from normal ones in high-resolution images.

To address these challenges, this article proposes a multilight-source computer vision method for fabric defect measurement, inspired by the significant variations in fabric defect features under different lighting conditions. The multilight-source image acquisition system includes a front reflection light source for determining camera exposure time and gain, a diffuse light source for basic defect measurement, a transmission light source for supplementary defect measurement, and a side light source to filter out nondefect regions, thereby improving measurement accuracy. Various fabric defect features are individually mapped into a salient feature space as effectively as possible, thereby enabling the decoupling of the complex measurement task through lighting. This strategy improves the accuracy of defect size measurement, significantly reduces inference time, and lowers algorithm development costs. To determine defect contours, a lightweight teacher–student architecture is designed to model the hypersurface of normal images under diffuse lighting during training, which is then used as a reference baseline during inference. To broaden the range of localizable defects, transmission light is employed to effectively capture common defect types that remain undetectable under diffuse lighting, and these defects are separated from the background through a trained model. During the training phase of the algorithm under transmission illumination, a self-supervised learning approach is adopted to simulate the shallow feature space of fabric defects. A lightweight segmentation network is then used to efficiently fit the semantic representation of these pseudo-defects in the deep feature space in real time. This method not only addresses the scarcity of defect samples but also enables rapid adaptation to different fabric types. To further improve the accuracy of defect size measurement, bidirectional side lighting is employed to distinguish nondefective regions, thereby achieving targeted separation of creases from the fabric via lighting. This eliminates the need for training a classification network and enhances adaptability across various fabrics. The main contributions of this work are as follows.

1) Introducing a size measurement framework for fabric defects under multiple illumination conditions, which is designed with an adaptive illumination control mechanism for dynamic adaptation to fabric variations. By leveraging diverse lighting, the complex measurement task is decomposed into three independent subtasks, significantly enhancing both measurement accuracy and inference efficiency.
2) Developing a lightweight teacher–student framework enables real-time defect localization under diffuse lighting. For transmission-light images, self-supervised learning generates pseudo-defects during training as high-quality inputs to a segmentation network, which compensates for the blind spots of diffuse lighting and improves localization accuracy.
3) Integrating the visual deviations caused by bidirectional side lighting on nondefect creases to resolve measurement inaccuracies typically introduced by such structures. Each subtask in the proposed framework supports real-time inference. Compared to existing methods, the proposed approach demonstrates greater practical value.

The remainder of this article is organized as follows. Section II reviews the relevant work in the field of fabric defect location based on machine vision. Section III presents the proposed measurement system. Section IV analyzes the experimental results. Finally, Section V concludes this article.

II. Related Works

The image-based approach analyzes fabric images captured by a digital imaging instrument to measure fabric defect

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

size. In this article, stabilized fabric images are dynamically acquired using a multilight-source image acquisition system, with the approach locating fabric defects under each light source for size measurement. The following is a review of the work related to the first step of defect size measurement, namely, fabric defect localization.

A. Fabric Image Acquisition Method

The task of defect size measurement consists of three steps: image acquisition, defect localization, and the calculation of the dimensions of the localized defects. The image acquisition system for defect localization comprises various acquisition tools, including high-resolution digital cameras, photoelectric sensors, and light-source devices [12], [13]. Imaging methods and illumination conditions are critical factors influencing the quality of input images. Therefore, existing approaches for defect size measurement employ multiple illumination conditions to improve image quality and enhance defect features, thereby overcoming the limitations associated with single lighting conditions [14], [15]. However, methods for measuring fabric defect size that rely on a single light source often suffer from missed detections and false localizations in practical applications, despite demonstrating good generalization performance for most prominent defect features [16], [17]. These issues significantly affect the localization of defect information, leading to high measurement errors. Even human inspectors adjust lighting conditions to carefully verify defect details. In other studies, although some defects can be localized, the high time costs and complexity of the algorithm reduce its feasibility for practical applications in production [5], [18]. In contrast, image acquisition systems utilizing multiple light sources can not only enhance defect feature representation by mapping them to a more distinguishable feature space, thereby reducing the complexity of real-time processing algorithms, but also facilitate multifrequency and multicategory measurements to improve accuracy. Because of the wide variety of defect types, the feature intensities under different light sources exhibit significant differences. More importantly, fabric, as a soft material, is prone to deformation, resulting in creases that disrupt the surface texture distribution and are easily mistaken for defects [19]. Considering the need for continuous monitoring of moving fabrics and for various defects to be efficiently captured, a multilight-source image acquisition system and a hierarchical detection process were proposed in our study.

B. Fabric Defect Location Algorithm

The complexity of defect data presents significant challenges for the location task, which can be addressed through traditional image processing or even deep learning-based techniques [20], [21], [22]. Supervised methods, such as the you only look once (YOLO) series, achieve high accuracy but require extensive annotations, limiting their practicality in defect detection where negative samples are scarce [23], [24], [25]. Unsupervised learning-based methods update network parameters from normal data to reduce the cost of data labeling, giving the model the ability to express normal paradigms. These methods include statistics-based methods, reconstruction-based methods, synthesis-based methods, feature embedding-based methods, and normalization flow-based methods. Statistics-based methods detect anomalies either by modeling the feature distributions of normal samples using multivariate Gaussian distributions or by identifying deviations from the modeled normal patterns [26], [27]. Reconstruction-based methods detect anomalies by comparing inputs with their reconstructions [28], [29]. Variational autoencoders (VAEs) and diffusion models have been used for this purpose, with diffusion models progressively denoising inputs to retain only normal features, making anomalous regions detectable through reconstruction discrepancies [30], [31], [32]. Synthesis-based methods generate artificial noise or anomalies that are used to enhance the data, improve discrimination, and reduce overfitting [33], [34], [35]. Feature embedding-based methods use pretrained networks to extract and compress normal features. These include memory banks and single-class classification methods to recognize anomalies through metric learning or predefined boundaries [36], [37]. Normalized flow-based models map normal data to Gaussian distributions to reduce the likelihood of anomalies [38], [39]. Knowledge distillation-based methods exploit differences between teacher and student networks for anomaly detection [40], [41], [42], [43]. Furthermore, a reverse distillation approach has been proposed for anomaly detection, achieving competitive performance [44], [45]. In this study, the majority of defects are localized through unsupervised learning by the proposed method, followed by comprehensive defect localization achieved by integrating the advantages of self-supervised learning with the goal-oriented characteristics of supervised learning. Moreover, critical challenges in the field of defect size measurement are addressed by employing a hybrid approach that combines both software and hardware techniques.

III. System

A. Fabric Image Acquisition System

Compared to defect localization methods based on a single light source, the proposed multilight-source defect image acquisition scheme effectively captures defect features, facilitating both defect localization and size measurement. In this system, the overall structure of the fabric defect size measurement system is presented, including the multi-light-source image acquisition setup, the localization algorithms corresponding to each light source, and the fusion mechanism used to determine the final defect size.

In a multilight-source image acquisition device, defects with complex characteristics may appear at various stages as the finished fabric undergoes multiple processing stages, making size measurement based solely on single-light-source conditions particularly challenging. To maximize the localization rate, an image acquisition method tailored has been developed, leveraging the substantial variations in defect feature distribution under different lighting conditions.

As shown in Fig. 2, the image acquisition system for full-roll fabric inspection comprises eight area-scan charge-coupled device cameras, a high-precision reflective mirror, a diffuse light source, a transmitted light source, and side-illumination sources. Fabric images under various lighting

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

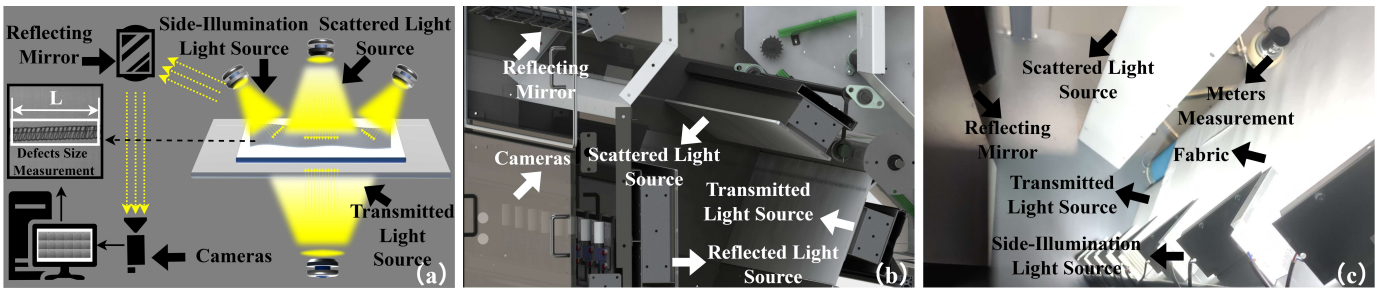


Fig. 2. Multilight-source alternating illumination enables comprehensive fabric image acquisition and effectively enhances defect features through light-source modulation, facilitating accurate size measurement. (a) Multilight-source image acquisition system, which includes orthogonal light (emitted from the front surface of the fabric), scattered light source, transmitted light source (emitted from the backside of the fabric), and side-illumination light source (illuminating the same fabric position from the top-left and top-right angles). (b) Three-dimensional virtual model of the image acquisition system, with a camera array that is uniformly aligned and housed in a sealed enclosure to prevent dust. (c) Actual image acquisition equipment is used to monitor product quality during inspection.

conditions are captured via the mirror and transmitted to the inspection system. All light sources utilize array-configured light-emitting diodes, with the diffuse source incorporating a diffuser to achieve uniform illumination. To ensure high-quality imaging and facilitate optimal lighting arrangement, area-scan cameras operating at 79 frames/s are horizontally mounted within a sealed enclosure to capture real-time fabric reflections. In practical image acquisition, a resolution of 2432 $\times$ 896 pixels is used, corresponding to a physical size of 28.9 $\times$ 10.7 cm. The power consumption of a single camera is less than 2.7 W at 50-V direct current (dc), and the signal-to-noise ratio (SNR) is 40.58 dB. The camera lenses are required to be dustproof, and the position of the mirror is adjustable to calibrate the illumination alignment. To achieve high-quality image acquisition and coordinated illumination, a multilight-source combination strategy is employed. Each light source consists of several LED panels. For front reflective illumination, diffuse illumination, and transmitted light, a total of five LED panels are used. Each panel operates at 48 V with a power of 160 W and has dimensions of 390 $\times$ 143 mm. In contrast, the side illumination uses smaller LED panels, each operating at 24 V with a power of 30 W and measuring 130 $\times$ 60 mm. These are arranged into eight groups, with two panels in each group. Different types of light sources respond distinctively to various defects, and the final image analysis is conducted via multithreaded parallel processing. First, the diffuse reflection light source illuminates the fabric at an approximate angle of 60°, which is determined empirically through engineering practice. Although this angle is not strictly fixed, it is chosen to avoid perpendicular incidence on the fabric surface since normal illumination tends to cause specular reflections that adversely affect defect localization. Using oblique diffuse lighting thus helps to capture texture and background information effectively while reducing image noise and surface glare. The imaging model is given by

$$I_{\text{diffuse}}(x,y) = R(x,y) \cdot L_d \cdot \cos(\theta_d) + \epsilon_d \tag{1}$$

where $R(x,y)$ denotes the reflectance of the fabric, L_d is the intensity of the diffuse light source, $\theta_d = 60°$, and ϵ_d is the sensor noise. The diffuse illumination provides uniform lighting, suitable for background modeling and overall illumination balancing. Next, the transmitted light source is directed vertically (90°) onto the fabric surface, primarily used to detect transmission-type defects such as oil. The image response model is expressed as

$$I_{\text{transmit}}(x,y) = T(x,y) \cdot L_t + \epsilon_t \tag{2}$$

where $T(x,y)$ represents the transmittance of the fabric at coordinate (x,y), L_t denotes the intensity of the transmitted light, and ϵ_t is the noise term. Slight oil contamination in the fabric, which locally alters the transmittance, results in noticeable bright or dark spots in the image. Finally, to reduce false positives caused by creases, two symmetrically arranged side light sources are introduced at angles of $\pm 60°$, resulting in a total angle of 120°. An angle of 120°, which typically falls within the range of oblique illumination, generates appropriate shadows and contrast on the fabric surface, thereby enhancing the visibility of texture details and minute defects and improving the image's depth and contrast. These side light sources enhance the directional contrast of creases in the images, aiding subsequent algorithms in crease identification. The model is described as

$$I_{\text{side},i}(x,y) = R(x,y) \cdot L_{s,i} \cdot \cos\left(\theta_{s,i}\right) + \epsilon_{s,i}, \quad i \in \{1,2\} \tag{3}$$

where $\theta_{s,1} = +60°$ and $\theta_{s,2} = -60°$ represent the incidence angles of the two side light sources. The exposure time for each light source is dynamically adjusted based on the fabric's advancing speed and the image acquisition frame rate, within a range of 0–300 μs. Let v denote the fabric speed and f denote the camera frame rate. The exposure time $t_{\text{exp},i}$ for the ith light source can be expressed as

$$t_{\text{exp},i} = \frac{d}{v \cdot f_i} \tag{4}$$

where d represents the desired pixel spacing on the fabric surface to maintain consistent spatial resolution and f_i is the effective frame rate or triggering frequency for the ith light source. Each acquired image from the different light sources is then processed in parallel by separate threads within a thread pool. Each thread applies a specific algorithm tailored to the characteristics of the corresponding illumination to accurately locate defects and measure their dimensions. The multithreaded processing strategy significantly enhances computational efficiency while ensuring high-precision defect size measurement.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

In the acquisition of fabric images, image brightness is primarily influenced by a combination of exposure time, camera gain, and the pulsewidth of the illumination source. Among these, exposure time and gain should be prioritized for adjustment to ensure the grayscale distribution of the image remains within a reasonable range. Specifically, longer exposure times result in brighter images; however, excessive exposure may lead to overexposure and detail loss. Conversely, overly short exposure times produce darker images, which hampers defect information extraction. Under the condition of satisfying brightness requirements, the camera gain should be kept as low as possible to reduce noise interference and prevent degradation of image quality caused by high gain. Once appropriate exposure time and gain values are determined, the brightness can be further refined by adjusting the pulsewidth of the light source. Pulsewidth defines the illumination duration; increasing it leads to higher image brightness. However, in high-speed imaging scenarios, excessively long pulse widths may introduce motion blur and reduce defect recognition accuracy. Therefore, the pulsewidth should be flexibly adjusted under the premise of maintaining image sharpness, serving as a secondary means of brightness control beyond exposure and gain.

Algorithm 1 Adaptive Brightness Adjustment Mechanism

Input: Initial exposure E_0, initial Pulsewidth Modulation(PWM) P_0, target brightness B_t, tolerance ϵ
Output: Exposure E, gain G, PWM P
Initialize $E \leftarrow E_0$, $P \leftarrow P_0$
Capture image, measure brightness B
if $B < B_t$ **then**
Binary search gain $G \in [0, G_{max}]$ until $|B - B_t| < \epsilon$
else if $B > B_t$ **then**
Binary search exposure $E \in [0, E_0]$ until $|B - B_t| < \epsilon$
end if
Lock camera parameters E, G
Adjust P for diffuse and side lights, keep E, G fixed
Adjust P for transmitted light to keep brightness in $[B_{min}, B_{max}]$

To accommodate varying fabric types and lighting conditions, this article proposes an adaptive illumination adjustment mechanism based on a binary search algorithm. Images captured under direct lighting are used as calibration References, and a grayscale value of $\text{Gray}_{\text{target}} = 120$ is set as the target brightness threshold to guide the overall image brightness into an optimal visually displayable range. The specific procedure is detailed in Algorithm 1. The fabric image captured under direct reflection illumination is first used as the reference for illumination adjustment, as it provides a more natural appearance and is visually more acceptable to the human eye. Subsequently, the camera's exposure time, gain, and the pulsewidth of the light source are initialized. The initial exposure time is empirically set to

$$T_{\text{exp}} = 300\ \mu\text{s}. \tag{5}$$

If the mean brightness of the image, denoted as $\text{Gray}_{\text{mean}}$, is lower than $\text{Gray}_{\text{target}}$, the gain G is iteratively increased using

$$G_{n+1} = G_n + \Delta G \quad (\Delta G > 0) \tag{6}$$

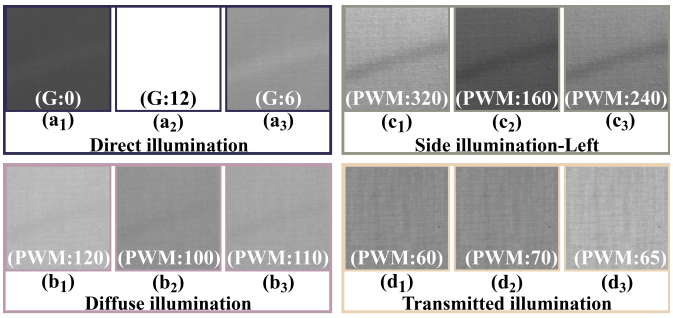


Fig. 3. Adaptive light regulation mechanism tailored to various textile materials. (a) Camera gain adjustment under direct reflective illumination. The exposure time and gain obtained from this illumination condition are used as reference values for the other light sources. (b)–(d) Grayscale variations of the captured images corresponding to pulsewidth adjustments under diffuse illumination, left-side illumination, and transmission illumination, respectively.

until

$$\text{Gray}_{\text{mean}} \geq \text{Gray}_{\text{target}}. \tag{7}$$

If $\text{Gray}_{\text{mean}} > \text{Gray}_{\text{target}}$ under the initial exposure time, a binary search is applied to iteratively reduce T_{exp}

$$T_{\text{exp}} = \frac{T_{\text{low}} + T_{\text{high}}}{2} \tag{8}$$

until the brightness falls within an acceptable tolerance

$$\left|\text{Gray}_{\text{mean}} - \text{Gray}_{\text{target}}\right| < \varepsilon. \tag{9}$$

Once the exposure time T_{exp} and gain G are determined during the direct illumination calibration stage, these parameters remain fixed during the entire imaging process to ensure the stability of the multi illumination system. After completing the direct illumination adjustment, rapid adaptation to diffuse and side-illumination sources, which exhibit brightness distributions similar to that of direct lighting, can be achieved by independently adjusting the pulsewidth modulation (PWM) of each light-source controller while maintaining fixed camera parameters. The brightness variations corresponding to different PWM settings for diffuse and side illumination are shown in Fig. 3(b) and (c). In Fig. 3(b), after fixing the camera exposure time and gain, the pulsewidth of the light source was adjusted using a binary search method to obtain high-quality images. Consequently, the pulsewidth value was decreased from 120 to 100 and then determined to be optimal at 110.

In contrast, transmitted illumination involves a more complex imaging path, as light penetrates the fabric material and exhibits different imaging characteristics. Therefore, independent PWM adjustment is also required. As shown in Fig. 3(d), the brightness target for transmitted illumination is set within the grayscale range from 150 to 180 since defects are clearly revealed within this range under transmitted light. This adjustment improves the identification and segmentation of fabric defects and supplements the detection of small defects that are difficult to observe under diffuse illumination.

B. Fabric Defect Location Algorithms

As shown in Fig. 4, the size measurement process consists of two parts. First, multilight-source localization is employed. Scattered light detects various defects, while transmitted light

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

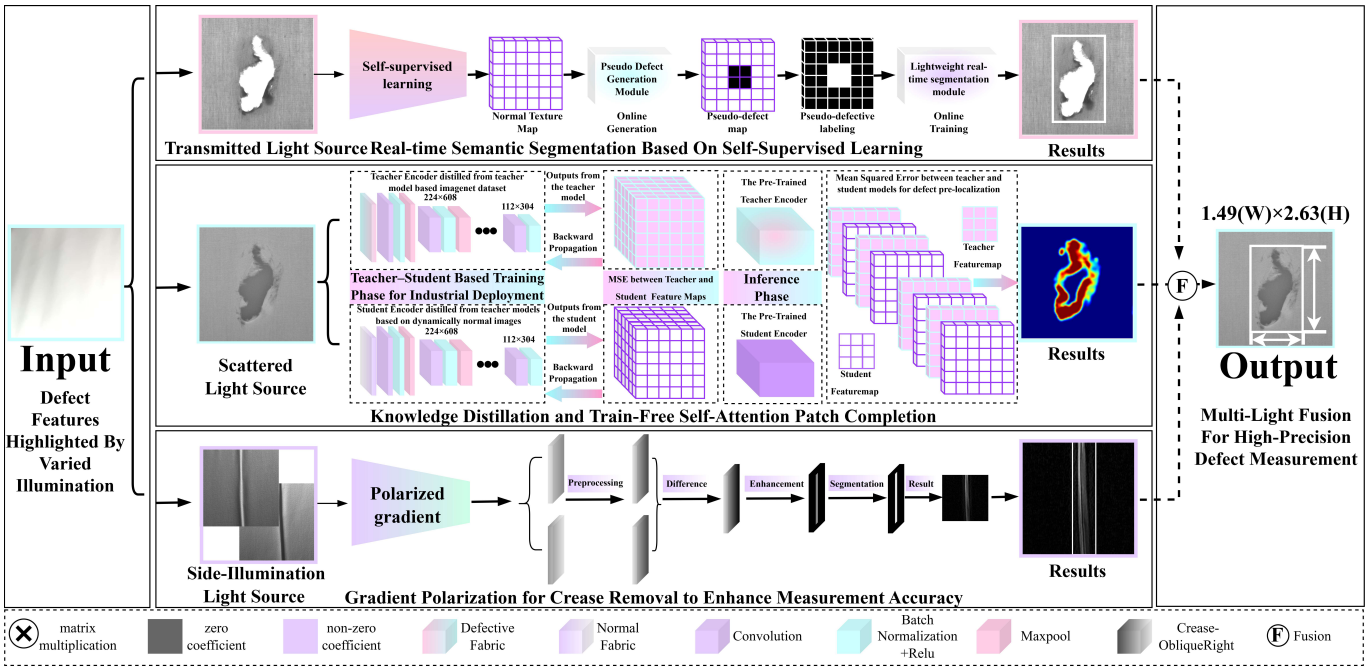


Fig. 4. This algorithm achieves defect size measurement in fabrics based on multilight-source image acquisition and a customized localization algorithm. The diffuse reflection light source is adopted as the primary processing stream due to its robustness to illumination variations and its ability to capture a greater number of defects, making it well-suited for visual analysis. A lightweight teacher–student structured unsupervised segmentation method is employed in the main thread to extract the majority of defect contours. In the auxiliary thread, considering the challenge of identifying oil stains under diffuse reflection, a transmitted light source is introduced to enhance their features. Pseudo-defect data are synthesized via self-supervised learning to augment the dataset, and semantic segmentation is then applied for precise oil stain detection. To address the issue of fabric creases being misclassified as defects in diffuse reflection images, a side-light source is used to highlight their distinct gradient characteristics. These features are identified through image processing techniques and matched with the primary thread results to improve accuracy. Finally, the outputs from the three light sources are fused to produce accurate defect contours, which are utilized for centimeter-level size measurement.

complements by identifying defects missed by scattered light; creases are then filtered out to reduce false positives. Second, results from multiple light sources are fused to determine the final defect bounding box for size measurement.

In the multilight-source defect location algorithm, compared to other light sources, scattered light effectively suppresses noise from surface textures and slight deformations, easing defect localization. Thus, it is crucial to develop a method that accurately represents the feature space of defective images and the normal sample hyperplane, maximizing their separation for efficient localization. These images form a dataset that exclusively consists of normal fabric samples, denoted as

$$Z = \{z_1, z_2, \ldots, z_N\}. \tag{10}$$

Given that fabric defects are rare events, this dataset is used to model the implicit feature subspace of normal fabrics, which serves as the reference for detecting unknown defective samples. Since most fabric defects exhibit significant differences from normal patterns in terms of texture, color, or structural regularity, we adopt the assumption that a test image q is likely to contain a defect if it deviates substantially from the feature distribution learned from normal samples. To learn the representation of normal fabrics and enable pixel-level defect localization, we adopt a teacher–student framework composed of two structurally identical convolutional neural networks: a teacher model $f_t(\cdot)$ and a student model $f_s(\cdot)$. To ensure absolute consistency in the nondefective regions of the feature space, both networks share the same architecture to ensure representational consistency in nondefective regions, but differ in their training strategy. Unlike traditional teacher–student frameworks, during training, we only guide the student to approximate the teacher's shallow features, allowing the student to quickly learn the normal paradigm. The teacher is distilled from a high-capacity feature extractor and remains fixed during the entire process. It provides reliable and stable feature representations. The student is trained only on the normal dataset Z by minimizing the pixel-wise mean squared error between its feature map and that of the teacher

$$\mathcal{L}_{\text{MSE}} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left\| f_s(z)_{i,j} - f_t(z)_{i,j} \right\|^2. \tag{11}$$

A lightweight teacher–student architecture is adopted, in which fabric images are encoded from 1216×448 pixels to 304×112 pixels through two stages of nonlinear transformation and downsampling. Inspired by Efficient-AD, only a single teacher–student pair is employed, as both logical and structural anomalies in fabrics are considered semantically equivalent for the purpose of this task. This simplification does not introduce significant computational overhead, thereby enabling real-time inference. Furthermore, hard example mining loss is incorporated to further improve defect localization accuracy. Through training, the student gradually approximates the teacher on normal data, thereby implicitly modeling the subspace of normal fabric features. At inference time, a query image q is first passed through both models. If the student feature $f_s(q)$ deviates significantly from the teacher feature $f_t(q)$, the discrepancy indicates potential anomalies.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

The distance between a query image and the normal set can be computed as

$$\text{Score}(q) = \min_{z \in Z} d\left(f_s(q), f_s(z)\right) \tag{12}$$

and the sample is classified as

$$\text{Category}(q) = \begin{cases} \text{Normal}, & \text{if Score}(q) < \tau \\ \text{Anomalous}, & \text{otherwise} \end{cases} \tag{13}$$

where τ is a predefined threshold. Furthermore, the pixel-wise difference between $f_s(q)$ and $f_t(q)$ can be visualized as a heatmap to localize defect regions at fine granularity.

Algorithm 2 Defect Segmentation Under Transmitted Illumination

Input: Transmitted image batch $\{I_i\}_{i=1}^N$
Output: Predicted segmentation masks $\{\hat{M}_i\}_{i=1}^N$
if Training Phase then
  for each image I_i in batch **do**
    $P_i \leftarrow \text{GeneratePseudoDefect}(I_i)$
    $M_i \leftarrow \text{GeneratePseudoMask}(P_i)$
    $\hat{M}_i \leftarrow \text{SegNet}(P_i)$
    $\mathcal{L}_i \leftarrow \text{CrossEntropyLoss}(\hat{M}_i, M_i)$
  end for
  $\mathcal{L} \leftarrow \frac{1}{N}\sum_{i=1}^N \mathcal{L}_i$
  Backpropagate $\mathcal{L}$ and update network parameters
else
  Testing Phase
  for each image I_i in batch **do**
    $\hat{M}_i \leftarrow \text{SegNet}(I_i)$
  end for
end if
Return $\{\hat{M}_i\}_{i=1}^N$

While excellent localization performance is exhibited by scattered light sources for certain defects, their inefficiency is revealed when dealing with light-sensitive defects that significantly affect fabric defect size. Using high-level features from deeper networks significantly increases temporal costs, limiting the feasibility of industrial applications. To address this issue, an innovative light-source design is proposed to map defect features that are difficult to localize under a scattered light source into a prominent feature space using a transmitted light source, thereby achieving cost reduction and efficiency enhancement. The characteristics of this type of illumination serve as a valuable supplement to diffuse reflection lighting, providing significant convenience for defect localization. However, due to the high visual similarity between such defects and the background, unsupervised methods face great challenges in addressing this task. Moreover, the limited number of defective samples makes it difficult to achieve satisfactory results using conventional supervised learning approaches.

Accordingly, an online training framework is developed, featuring pseudo-defect synthesis for transmitted light images and seamless compatibility with diffuse reflection imaging. The proposed method comprises two main components: an online pseudo-defect generation module and a real-time defect segmentation module. As illustrated in Algorithm 2, the process begins by analyzing common characteristics of actual

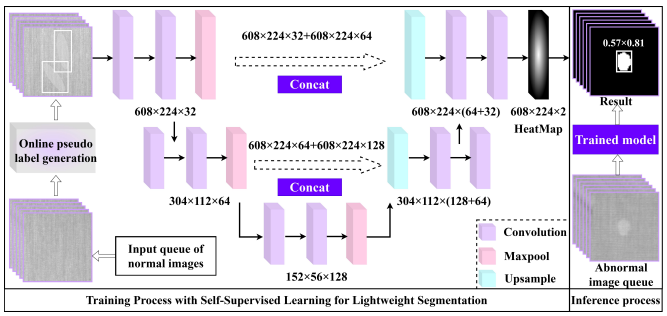


Fig. 5. Lightweight UNet-based semantic segmentation network is specifically designed for oil stain detection. During the training phase, normal samples are first used as the input image queue, and pseudo-defects are generated through a pseudo-label generation module. These augmented images are then fed into a lightweight U-shaped network for oil stain semantic segmentation. In the testing phase, the input image queue is directly used for oil stain detection. Since the model is trained in a supervised manner, it is sensitive only to oil stain defects, thereby enabling targeted localization.

defects, particularly their similar pixel intensity distributions. The shape features of such defects are often constrained within a specific subspace, which serves as the foundation for pseudo-defect generation. In contrast to conventional methods, the defects of interest are typically composed of elliptical-like structures with pixel intensities closely resembling the background. Introducing nontargeted defect features, such as those generated by advanced methods like Perlin noise, may potentially degrade the performance of the network. To address this, transmitted light is employed as a complementary illumination source, enabling the generation of defect features that are spatially and semantically aligned with actual defect types. The pseudo-defect is modeled as an elliptical region on the fabric image, defined by

$$\frac{(x-c_x)^2}{a^2} + \frac{(y-c_y)^2}{b^2} \le 1 \tag{14}$$

where (c_x, c_y) is the defect center and a and b are the ellipse's semi-axes. The pixel intensity inside this region is blended with the original image as

$$I'(x,y) = (1-\alpha)\, I(x,y) + \alpha I_d \tag{15}$$

where $I(x,y)$ is the original pixel value, I_d is the defect intensity, and α controls the blending strength. The corresponding mask is

$$M(x,y) = \begin{cases} 1, & \text{if } \dfrac{(x-c_x)^2}{a^2} + \dfrac{(y-c_y)^2}{b^2} \le 1 \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

$$L(x,y) = I'(x,y) \cdot M(x,y). \tag{17}$$

This ensures pseudo-defect mimic real defect shapes and intensities under transmitted light. The locations and sizes of these pseudo-defects are modeled as random variables to simulate realistic conditions, as shown in Fig. 5. By adopting this strategy, the original discrete defect detection task is reformulated as a targeted segmentation problem, which is better suited to the capabilities of supervised learning. To ensure accurate defect segmentation while maintaining high efficiency, a lightweight segmentation network is designed based on a U-shaped architecture. This network effectively

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

distinguishes generated prior knowledge from the background and achieves stable, low-loss convergence within minutes of training, thereby demonstrating strong performance in both robustness and computational efficiency.

The defects on the fabric surface are localized under scattered light sources due to varying degrees of deformation. However, creases that do not belong to defects are the primary cause of false positives. Therefore, distinguishing crease regions is essential for reducing false positives. Inspired by the significant changes in the pixel gradient of the crease region, the fabric images under both lighting conditions are first denoised and the fabric texture is weakened using

$$I' = I * G \tag{18}$$

$$G(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{19}$$

where I denotes the input image and x and y represent the horizontal and vertical coordinates within the Gaussian kernel (filter), respectively. σ is the standard deviation, with a value of 0.8. Here, * denotes the convolution operation, which is applied to achieve noise reduction while preserving crease details. Due to the two sets of light sources positioned at a 120° angle illuminating the left and right sides of the crease, the weakened crease features are mapped onto a gradient-polarized feature space, which significantly reduces the algorithm's complexity and time cost. Therefore, the difference between the two filtered images is computed separately to allow for the preliminary identification of the crease regions as quickly as possible. It can also be calculated as follows:

$$D_{\text{iff}}(x,y) = \left|I'_1(x,y) - I'_2(x,y)\right| \tag{20}$$

$$I_3(x,y) = \text{Otsu}\left(D_{\text{iff}}(x,y)\right) \tag{21}$$

$$I_4(x,y) = \text{M}\left(T(x,y)\right) \tag{22}$$

where I'_1 and I'_2 denote the images after filtering under a side-illumination light source. Since the two images exhibit opposite gradient distributions at the same location, a difference operation is first applied to obtain D_{iff}. This step not only reduces computational complexity but also enables precise identification of crease regions. Next, features in the horizontal direction are extracted to emphasize the crease regions, while areas with significant gradient variations in the vertical direction are associated with another type of defect. Finally, the *Otsu* method is employed to preliminarily identify the crease regions I_3, which are further refined using morphological operations M to obtain I_4, enhancing the accuracy and effectiveness of localization.

C. Size Measurement Through Multilight-Source Integration

As shown in Fig. 6, each light source detects defects based on their characteristic differences, thereby enhancing localization precision and measurement accuracy. As shown in Algorithm 3, the result under scattered light I_{diffuse} serves as the primary reference for defect localization, while the images under transmitted light I_{trans} and side light I_{side} are used as auxiliary references to maximize measurement accuracy. On the one hand, if oil stain defects are localized under the transmitted light source, they are stored in the defect result queue to improve measurement accuracy. On the other hand,

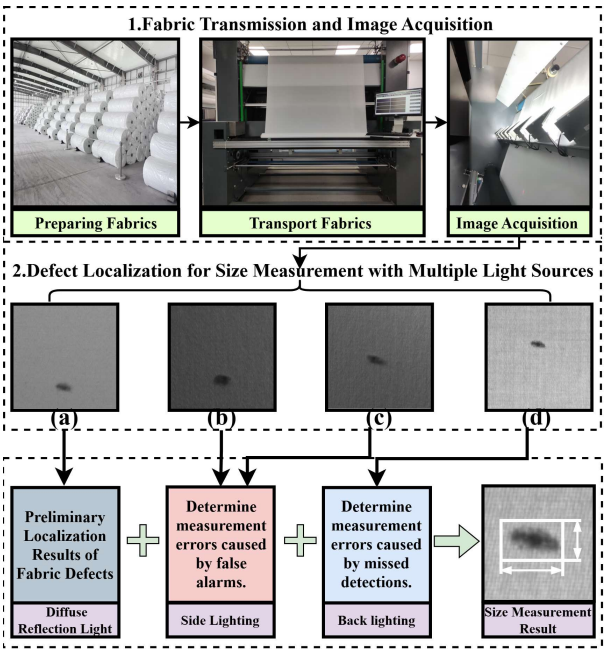


Fig. 6. Measurement of fabric defect size consists of two main steps. (a) Fabric defect imaging under a scattered light source. (b) and (c) Fabric defect imaging under a side-illumination light source. (d) Fabric defect imaging under a transmitted light source. First, images are captured and stored using multiple light sources. Subsequently, the terminal analyzes the images in separate thread pools based on the light source type, locates the defects, and calculates their minimum bounding rectangles to determine the defect dimensions in the warp and weft directions.

if crease defects are detected under the side light source, the corresponding regions are excluded from size measurement by freezing the crease areas under the scattered light source based on pixel position offsets. Finally, the localization results from multiple illumination sources are integrated for comprehensive defect size measurement.

Algorithm 3 Defect Detection and Dimensional Measurement Through Multi Illumination Fusion

Input: Diffuse image I_{diffuse}, transmitted image I_{trans}, side illumination image I_{side}
Output: Final defect measurements M
$D_{\text{diffuse}} \leftarrow$ DetectDefects(I_{diffuse})
$D_{\text{trans}} \leftarrow$ DetectTargetDefects(I_{trans})
$C_{\text{side}} \leftarrow$ DetectCreaseContours(I_{side})
Initialize result set $M \leftarrow \emptyset$
for each defect $d \in D_{\text{diffuse}}$ **do**
  $contour \leftarrow$ SegmentationContour(d)
  if Overlap($contour, C_{\text{side}}$) = False **then**
    $M \leftarrow M \cup \{\text{Measure}(contour)\}$
  end if
end for
for each defect $d \in D_{\text{trans}}$ **do**
  $contour \leftarrow$ SegmentationContour(d)
  $M \leftarrow M \cup \{\text{Measure}(contour)\}$
end for
Return M

Size measurement is conducted by determining defect contours and calculating their minimum bounding rectangles.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

Since fabric defects typically occupy a small area in the image, pixel-level metrics are considered inadequate for comprehensively evaluating model performance. Therefore, an evaluation metric based on instance-level matching and size error is proposed in this study. Minimum bounding rectangles are extracted from the binary maps of predicted and ground-truth defects to form the sets of predicted and ground-truth boxes. A greedy matching algorithm based on intersection over union (IoU) is employed, where candidate box pairs with IoU values exceeding a predefined threshold are matched in descending order of IoU to establish one-to-one correspondences [46]. The number of matched pairs is defined as true positives (TPs), while unmatched ground-truth and predicted boxes are categorized as false negatives (FNs) and false positives (FPs), respectively. For each matched box pair, the relative error of the longer side is calculated, whereas a fixed error value of 1 is assigned to unmatched boxes. The final size error is computed as the weighted average of all errors, normalized by the total number of TP, FP, and FN. A smaller size error is interpreted as an indication of higher prediction accuracy and better model performance.

IV. Experiments and Discussion

A. Dataset and Evaluation Metrics

To thoroughly validate the practical applicability of the proposed system, the dataset was collected from a real-world textile manufacturing environment. The dataset is divided into two parts: a training set with 6000 normal fabric images and a test set with 508 defective images covering nine types of fabric defects, including oil-stained warp, floating fibers, grease spot, dirt spot, thick weft, hole, oil dot, broken warp, and joint

$$E_\epsilon = \frac{\left|P_{\text{Length}} - T_{\text{Length}}\right|}{P_{\text{Length}} + T_{\text{Length}}} \tag{23}$$

$$E_l = \eta \cdot \text{FP} + \eta \cdot \text{FN} \tag{24}$$

$$\Omega = \frac{E_\epsilon + E_l}{N} \tag{25}$$

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{26}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{27}$$

$$F1 = \frac{2 \cdot P \cdot R}{P + R} \tag{28}$$

$$\text{Throughput} = \frac{N_{\text{images}}}{T_{\text{seconds}}} \tag{29}$$

$$\text{Speed} = \frac{\text{Throughput} \times \text{Unit Length} \times 60}{N_{\text{Cameras}}}. \tag{30}$$

To evaluate the measurement performance of the proposed method as well as to satisfy the deployment requirements for practical application scenarios, several metrics are adopted. The size measurement error E_ϵ is defined as the normalized absolute difference between the predicted length P_{Length} and the true length T_{Length}, where normalization is performed by dividing by their sum. The smaller the value of E_ϵ, the lower the measurement error, and the higher the measurement accuracy. The measurement error is computed based on the matched pairs of ground-truth and predicted bounding boxes, while unmatched boxes are regarded as false negatives and false positives. However, computing the measurement error only for successfully matched instances cannot reflect the overall performance of the model since accurate measurement relies on correct localization. To quantitatively evaluate the impact of unmatched instances on the overall performance, an error penalty factor η is introduced and applied to both false positives and false negatives, yielding $\eta \cdot \text{FP}$ and $\eta \cdot \text{FN}$. In our experiments, η is set to 1, and their sum is defined as E_l, representing the maximum contribution to the error. To evaluate whether the proposed method achieves both accurate localization and precise measurement, the comprehensive defect localization coefficient Ω is proposed. It reflects not only whether defects are correctly localized but also whether the size of successfully localized defects is accurately measured. For normalization, the measurement errors of all successfully matched defects, together with the penalty errors of unmatched defects, are summed and expressed as a percentage relative to the total number of defect instances N since the error is computed on a per-instance basis. This formulation demonstrates the comprehensive performance of the proposed method. P represents the proportion of true defects among all samples identified as defective. R refers to the proportion of actual defect samples that are correctly detected as defects. $F1$-score measures the model's ability to maintain high precision while effectively capturing the majority of true anomalies. In addition, timeliness metrics such as latency, throughput, and processing speed are critical factors in assessing the feasibility of practical applications, particularly in fabric defect size measurement. Latency refers to the time required to process a single image. Throughput indicates the number of images N_{images} processed per second, given by $(N_{\text{images}}/T_{\text{seconds}})$. Speed refers to the real-time measurement rate in meters per minute, where the *unit length* represents the physical length of the fabric in the longitudinal direction corresponding to 896 pixels, and $N_{Cameras}$ indicates the number of cameras, which is 8, covering the entire width of the fabric.

B. Implementation Details

In the location phase, computations under scattered light sources are conducted using PyTorch 2.1.1. All of the models use an NVIDIA RTX3090 GPU for training and testing. The Adam optimizer is used for training with the cosine lr schedule. The long side of the image is 2432 pixels, which leads to the batch size being only 4. Training proceeded for 300 epochs. For deployment, the model is optimized using TensorRT 1.16.1, which is implemented in C++ to further enhance the location speed. The algorithms for transmitted light source and side-illumination light-source conditions are executed on a CPU (Intel i9-9900K).

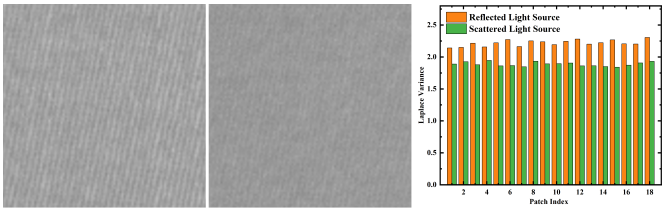
C. Baseline Light-Source Validation

To validate the effectiveness of each light source, defect features under different illumination conditions were compared, and the defect localization results from various algorithms were visualized to assess their respective contributions to size measurement accuracy. Fig. 7(a) and (b) shows that textures under specular reflection are clearer than under scattered light. The Laplacian variance of corresponding image blocks,

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

TABLE I
VERIFICATION OF THE EFFECTIVENESS OF EACH LIGHT SOURCE

Light Source	Ω	P	R	F1
Scattered Light	37.35	89.96	84.01	86.75
+Transmitted Light	25.58	90.20	97.61	93.77
+Side-Illumination Light	21.59	94.99	97.61	96.24



(a) (b) (c)

Fig. 7. (a) Visibility of the texture under reflected light. (b) Effective suppression of texture by scattered light source. (c) Quantitative analysis of the difference in texture prominence under the two light sources.

quantified in Fig. 7(c), is consistently lower under scattered light, indicating effective texture suppression [47]. As shown in Fig. 8, defect size measurement under scattered light is performed by first generating a heatmap via knowledge distillation (b), followed by threshold-based segmentation (c). Morphological operations connect adjacent regions, and the minimum bounding rectangle is used to measure defect length and width, enabling size error calculation. As shown in Table I, when the scattered light source is used as the sole illumination, the recall and $F1$-score are limited to 84.01% and 86.75%, respectively. This results in a value of $\boldsymbol{\Omega}$ of 0.37, highlighting the inadequacy of relying solely on scattered light for accurate defect size estimation. Consequently, incorporating additional light sources becomes essential to reduce false negatives and false positives, thereby improving measurement precision and overall system robustness.

D. Effectiveness Validation of Auxiliary Illumination

To address the limitations of diffuse illumination in detecting oil contamination, feature representations of oil-stained regions under both diffuse and transmitted lighting conditions were visualized and compared, as shown in the figure. Under diffuse illumination, the oil-contaminated areas exhibit low visual contrast, making their detection challenging. Although the features under transmitted illumination appear more prominent, they still exhibit visual similarity to the background. To tackle this issue, a supervised segmentation method was developed, by which oil-stained pixels are accurately separated from the background. Fig 9(a) and (b) shows that the defect is nearly indistinguishable under diffuse illumination, which complicates defect detection. To address this, transmitted light images in Fig. 9(c) and (d) are used, where oil stains appear clearly. To localize such defects, the proposed self-supervised learning method was trained, with its loss and $F1$-score shown in Fig. 10, demonstrating effective convergence and high accuracy. Finally, size measurement results of oil stains under transmitted light are presented in Fig. 11 to further validate the localization performance. As a result, the recall rate increases

TABLE II
REAL-TIME INFERENCE VALIDATION OF THE SYSTEM

Light Source	Latency (ms/img)	Throughput (img/s)	Speed (m/min)
Transmitted Light	6.02	166.05	133.22
+Scattered Light	23.78	42.04	33.71
+Side-Illumination Light	32.26	31.00	24.88

to 97.61% under transmitted illumination, further reducing the value of $\boldsymbol{\Omega}$ to 0.25.

The proposed method combines multiple light sources and multiple algorithms to achieve full recall of defects with irregular textures or color differences as well as feature differences. Nonetheless, crease features are often susceptible to false positives, compromising measurement accuracy. Under scattered light sources, weak crease features are particularly prone to being misclassified as defects by the algorithm, leading to an elevated false positive rate. By employing a side-illumination light source that emits light in lateral directions, the gradient characteristics of creases are effectively enhanced. After obtaining the crease image enhanced by the multilight-source acquisition system, we visualized the pixel gradient differences of the crease features before and after enhancement under the side-illumination light source. Fig. 12(a) shows that under side-illumination light source, the pixel gradient of the crease features is at least three times greater than under a scattered light source. Therefore, the method based on gradient discontinuities for detecting creases is effective. As illustrated in Fig. 12(b), crease localization under side-illumination light source demonstrates strong generalizability, while the presence of side-illumination light source reduces the value of $\boldsymbol{\Omega}$ to 0.21.

E. Real-Time Inference Validation of the System

As shown in Table II, to validate the stability of multiple algorithms under varying illumination conditions, the inference time of each algorithm was systematically evaluated. Given the variations in execution time under different lighting sources, we analyzed the inference latency for each illumination in ascending order, as well as the total system latency, to determine whether the fabric inspection process meets real-time requirements. First, images under transmission illumination are primarily used for detecting oil stains, which are typically localized defects. To enhance processing efficiency, these images were downsampled to a resolution of 608×224 pixels. Combined with the use of a lightweight segmentation network, the inference time under transmission illumination is significantly reduced. However, relying solely on transmission illumination is insufficient. Under diffuse illumination, which serves as the primary detection thread to ensure high measurement accuracy, the algorithm exhibits notably longer inference times due to the increased complexity and comprehensive detection requirements. In contrast, side illumination mainly enhances the visibility of fabric creases by manipulating lighting rather than relying on deep models. Traditional image processing techniques are employed under

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

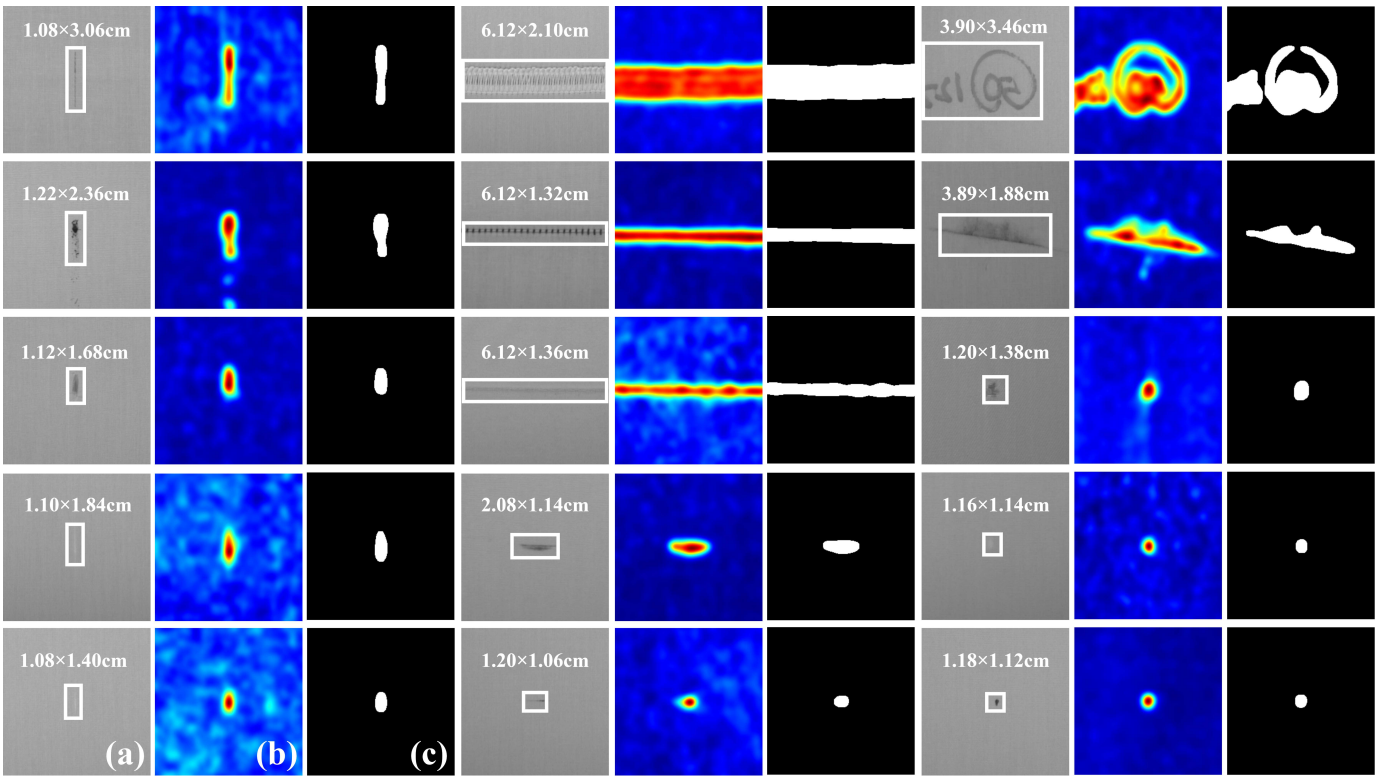


Fig. 8. Fabric defect size measurement results under a scattered light source. (a) Original image with defect contours and the minimum enclosing rectangle for measuring the defect width and length, with numerical values representing the real dimensions in centimeters. (b) Heatmap output by the neural network. (c) Defect segmentation result used for measurement.

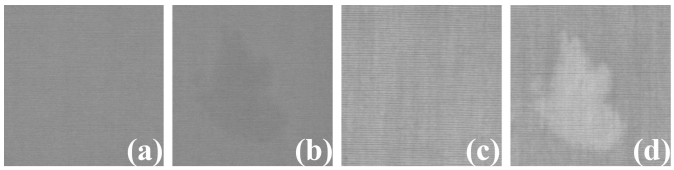


Fig. 9. Defects that cannot be localized under scattered light exhibit relatively distinct features under transmitted light. (a) Normal image under scattered light source. (b) Oil stain defects under scattered light show weak features, making localization difficult. (c) Normal image under transmitted light source. (d) Enhanced oil stain features under transmitted light facilitate accurate contour determination.

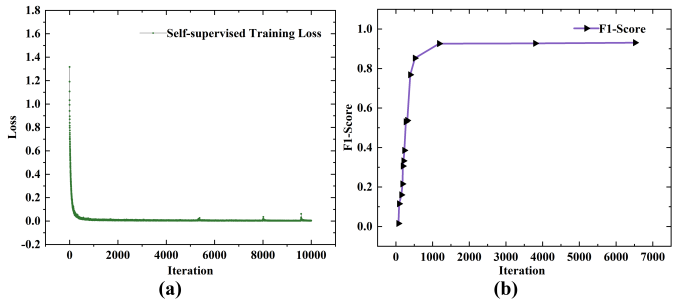


Fig. 10. Training loss curve under self-supervised learning and the $F1$-score curve evaluated every ten iterations are presented. (a) Pseudo-label generation loss demonstrates convergence. (b) $F1$-score increases as training proceeds.

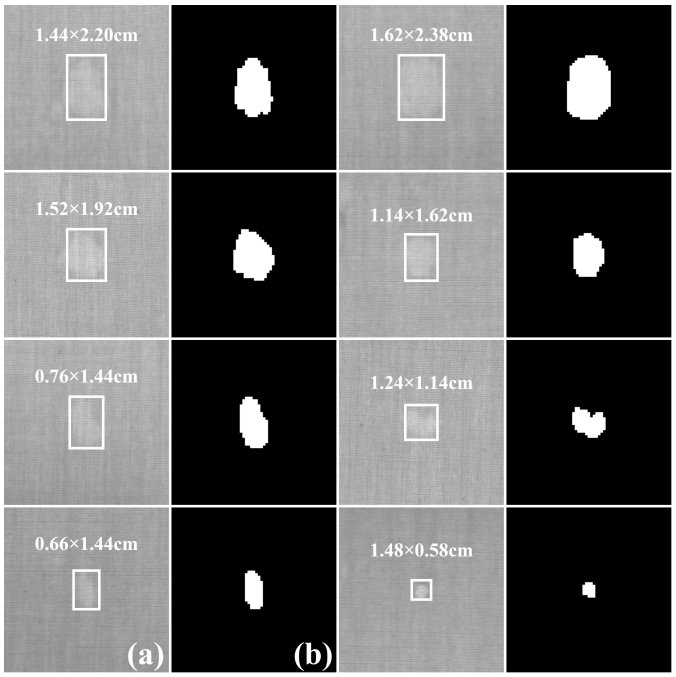


Fig. 11. Semantic segmentation results of oil stain defects and contour-based size measurements using the proposed pseudo-label-assisted framework. (a) Size measurement of the oil stain defect based on localization contours, presented in the form of width × height. (b) Semantic segmentation result of the oil stain defect, facilitating accurate contour-based size estimation.

this condition to identify and exclude crease regions, thereby reducing false positives and improving the reliability of size estimation. Ultimately, the proposed size measurement system achieves an inference time of 32 ms per image, corresponding to a throughput of 31 frames/s. Considering the pixel-to-physical scale of 0.12 mm per pixel and the parallel image acquisition by eight cameras, the system reaches an overall

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE COMPARISON OF MEASUREMENT RESULTS USING VARIOUS SOTA METHODS

Method	E_ϵ	Ω	P	R	F1	Latency (ms)	Throughput (img/s)	Speed (m/min)	Model Size MB	GPU Memory Usage GB	Parameters MB
Dinomaly	26.24	50.17	68.31	78.43	73.02	135.88	7.36	5.91	546.61	2.33	148.01
INP-Former	32.45	67.60	54.06	64.53	58.83	132.66	7.54	6.05	591.67	2.44	155.10
UniNet	34.62	64.55	56.56	74.04	64.13	166.35	6.01	4.81	585.42	6.57	153.46
RD4AD	24.80	46.56	74.41	74.95	74.68	46.18	21.66	17.38	357.82	1.81	93.80
Efficient-AD	22.33	40.00	75.94	81.50	78.62	34.81	28.73	23.06	30.77	1.85	8.07
RD++	38.13	48.28	85.80	77.33	81.35	53.52	18.68	14.99	416.90	1.49	109.29
FCA	25.10	35.96	82.55	83.15	82.85	559.63	1.79	1.43	**15.74**	6.28	**4.13**
FastFlow	42.42	73.85	52.95	60.69	56.56	140.35	7.13	5.72	476.53	13.98	124.92
DDAD	26.84	77.31	35.90	55.39	43.57	3179.58	0.31	0.25	357.82	19.42	93.80
AST	30.31	49.83	71.18	80.95	75.75	53.23	18.79	15.08	323.00	1.68	84.67
CFA	25.01	37.79	80.46	82.82	81.62	88.91	11.25	9.03	42.64	0.58	11.18
STFPM	30.47	48.65	74.35	78.98	76.60	50.30	19.88	15.96	89.18	0.94	23.38
Patchcore	23.86	41.78	78.76	76.74	77.74	169.80	5.88	4.82	262.77	**0.41**	68.88
Ours	**16.02**	**21.59**	**94.99**	**97.61**	**96.24**	**32.26**	**31.00**	**24.88**	20.56	0.92	5.38

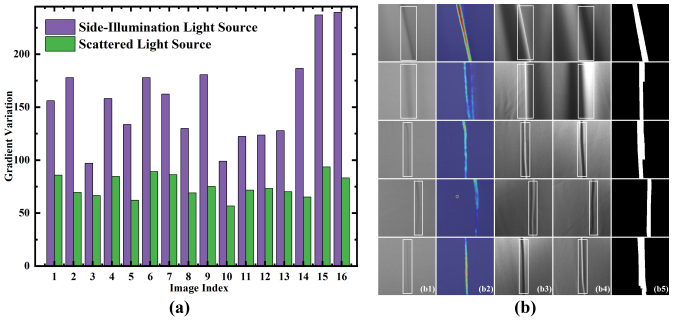


Fig. 12. Comparison of image gradient enhancement and crease detection under scattered light source and side-illumination light source. (a) Before enhancement, pixel gradient fluctuations in crease regions remain steady; after enhancement, they become more pronounced. (b) Creases under scattered light source are often mistaken for defects, leading to measurement errors and inaccurate fabric grading. (b1) Crease image under scattered light source. (b2) Defect heatmap under scattered light source. (b3) and (b4) Crease images under side lighting. (b5) Crease localization under side lighting.

inspection speed of 25 m/min, outperforming manual inspection in both speed and operational efficiency. The inference latency of each processing thread, along with the aggregated system delay, confirms that the proposed system delivers stable and efficient performance, effectively satisfying the real-time requirements for fabric defect detection and dimensional measurement.

F. Comparison With Existing Methods

To fully validate the advantages of the proposed method over existing approaches, quantitative comparative experiments were conducted, focusing on measurement accuracy and timeliness. The evaluated methods span a wide range of techniques, including state-of-the-art (SOTA) deep learning models such as Dinomaly [29], INP-Former [28], and UniNet [40]; statistics-based methods such as CFA [27]; memory bank-based methods such as PatchCore [36]; knowledge distillation-based approaches such as Efficient-AD [41], RD4AD [45], RD++ [44], AST [42], and STFPM [43]; normalizing flow-based methods such as FastFlow [39]; diffusion model-based methods such as DDAD [32]; and zero-shot approaches such as FCA [48]. As shown in Table III, the proposed method achieves a measurement error of 0.16, which is lower than all compared approaches, demonstrating its high measurement accuracy. The combined metric for both measurement and localization also outperforms SOTA methods, indicating that the proposed approach not only minimizes measurement error but also excels in defect localization. To further evaluate performance at the individual defect level, we applied the proposed instance-level matching rules to compute precision and industrial-grade recall for each defect, achieving 94.99% and 97.61%, respectively, with an $F1$-score of 96.24%. The multi illumination-based method effectively reduces both false negatives and false positives, enabling comprehensive defect recall that even SOTA deep learning methods cannot achieve for all instances. Moreover, for defects such as creases, the designed illumination strategy not only reduces algorithm development costs for such pseudo-defects but also saves computational time and memory, which is critical in industrial environments where prolonged GPU operation can generate excessive heat and complicate cooling. To further demonstrate the industrial feasibility of the proposed method, we evaluated its efficiency. The method achieves a per-frame processing time corresponding to an inspection speed of 24.88 m/min, exceeding that of manual inspection. Comparative experiments confirm that the proposed approach meets industrial timeliness requirements. Since the current implementation uses a single host and already exceeds human inspection speed, adding additional hosts would further increase inspection throughput.

To demonstrate the robustness of the proposed method, quantitative experiments were conducted to assess the size

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

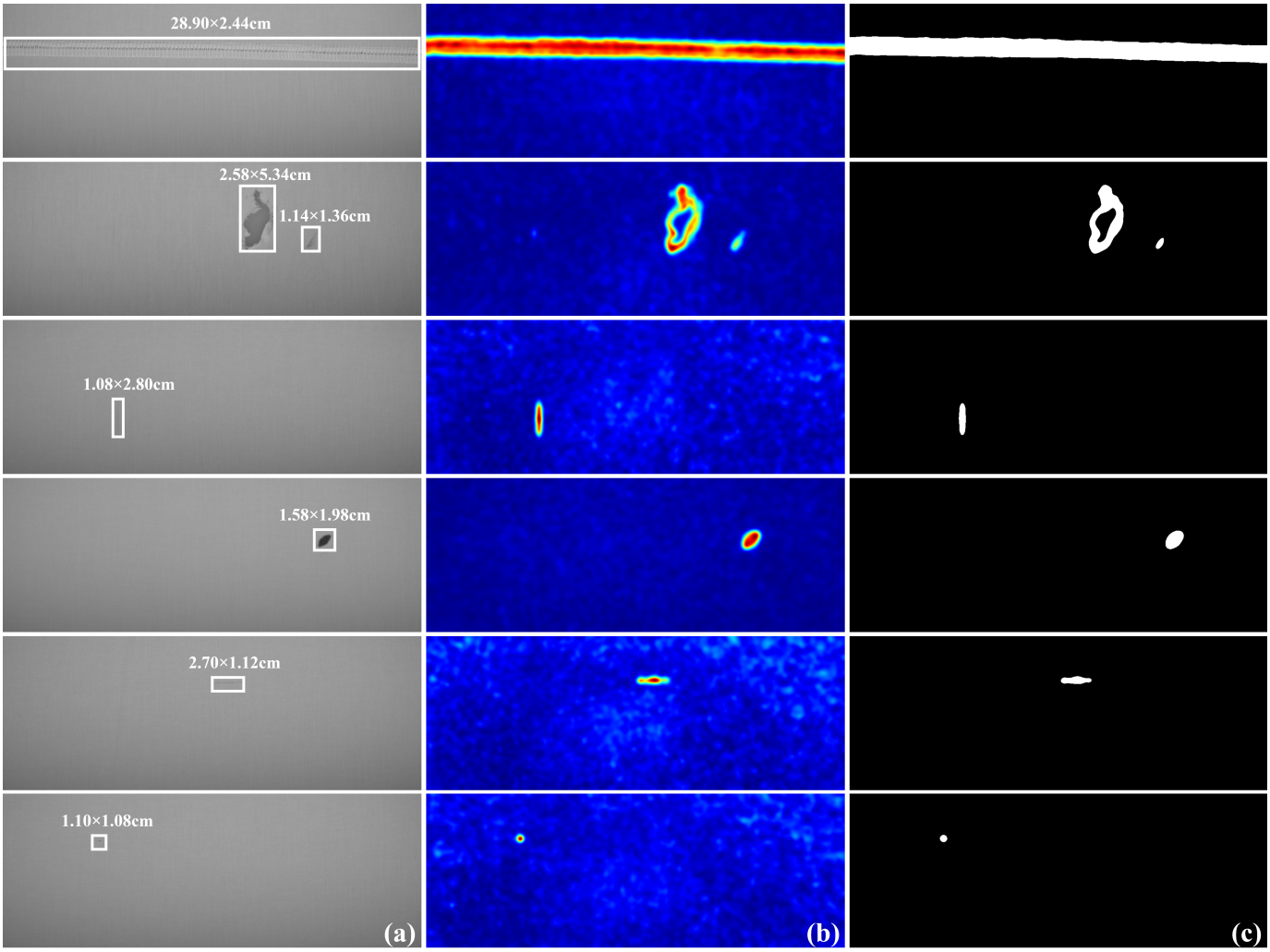


Fig. 13. Measurement results for grayscale images with a resolution of 2432 × 896 pixels conform to industrial standards. (a) Defective images captured under scattered light sources, with bounding boxes (width × height) indicating the locations of defects. (b) Corresponding heatmaps highlighting defect regions. (c) Fabric defect segmentation maps obtained from the proposed method.

TABLE IV

QUANTITATIVE COMPARISON OF DIFFERENT SOTA METHODS ON MEASUREMENT ERROR $\mathbf{E}_\epsilon$ FOR MULTIPLE DEFECT CATEGORIES

Model/Defect	Oil-stained Warp	Floating Fibers	Grease Spot	Dirt Spot	Thick Weft	Hole	Oil Dot	Broken Warp	Joint
Dinomaly	23.32	21.93	33.54	28.80	31.84	27.50	23.65	24.95	29.07
INP-Former	41.17	22.26	30.91	24.24	26.38	36.60	38.70	25.89	35.22
UniNet	45.38	23.91	25.40	33.77	28.86	32.93	50.27	31.01	29.40
RD4AD	35.76	9.40	14.06	34.57	28.68	23.77	27.57	11.55	25.61
Efficient-AD	22.11	19.75	18.49	22.46	27.88	24.34	24.93	18.71	27.79
RD++	46.93	23.38	24.65	37.77	**46.97**	43.74	48.03	23.78	34.39
FCA	33.20	12.58	13.12	30.27	30.54	29.48	35.57	8.90	16.96
FastFlow	**68.58**	23.94	31.53	**46.96**	36.86	**47.32**	**63.65**	24.70	32.03
DDAD	30.60	20.00	28.12	24.68	28.83	25.86	37.34	22.61	26.36
AST	25.02	29.54	35.35	23.87	37.29	24.23	35.09	32.60	**50.66**
CFA	14.63	**36.99**	**38.75**	20.98	20.03	19.14	23.08	**37.86**	37.62
STFPM	44.75	16.04	19.09	27.71	25.51	28.22	43.67	19.68	25.83
Patchcore	28.05	16.34	23.43	19.50	21.32	22.66	28.47	19.42	29.21
Ours	**19.13**	**9.90**	**13.66**	**17.14**	**17.52**	**20.29**	**14.96**	**10.31**	**15.24**

measurement errors $\mathbf{E}_\epsilon$ for multiple defect types, as presented in Table IV. The results show that the proposed approach achieves a minimum error of 0.09 and a maximum error of 0.20 across individual categories. Although some categories

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

exhibit lower measurement errors, the differences are marginal, and it is essential to balance the overall performance for industrial applications. Since the method demonstrates consistent error levels across multiple categories without extreme deviations, its robustness and practicality are validated through quantitative experiments against various SOTA methods. In addition to quantitative experiments, we also conducted quantitative visualization to assess measurement performance under realistic industrial conditions. Unlike other materials, fabrics cover large areas with widely distributed defects, exhibiting multiscale shapes and size variations. This necessitates rigorous defect search across the entire fabric surface to ensure high-quality assessment. To illustrate the performance of the proposed defect measurement method on original images, we visualized the defects at a resolution of 2432×896 pixels, including the original images, corresponding heatmaps, and binary maps, as shown in Fig. 13. The visualizations confirm that the proposed method maintains strong robustness and generalization capability across defects of various scales.

V. Conclusion

Fabric defect size is a key factor in quality evaluation. Current methods using single lighting fail to measure size accurately and suffer from false positives and negatives, causing significant errors. This article proposes a multilight-source approach that simulates and surpasses human multiangle inspection to achieve precise defect size measurement and decouple complex quality issues. Unique features under each lighting are analyzed, and real-time localization algorithms are adapted to improve measurement accuracy. First, a set of scattered light sources is employed for basic defect localization, based on a lightweight teacher–student framework. Then, transmitted light sources are used to compensate for missed detections, guided by a proposed pseudo-label generation strategy that combines the strengths of supervised and self-supervised learning, effectively reducing the missed detection rate. Finally, defect localization results are integrated based on the complementary effects of multiple light sources, producing a consolidated output with accurate bounding boxes to calculate the error between the predicted defect size and the true size. Experimental results demonstrate that the proposed multilight-source and multialgorithm collaborative approach achieves superior performance, with precision and recall rates of 94.99% and 97.61%, respectively, surpassing existing methods. Moreover, the method attains a high $F1$-score, effectively enhancing the accuracy of size measurement and reducing the measurement error to 0.16, thereby significantly improving the reliability and robustness of defect detection. The system operates in real time, surpassing manual inspections. It meets industrial fabric quality assessment requirements and enhances the daily workload capacity for enterprises in product quality control.

With the continuous growth of textile enterprises and the increasing daily output of fabrics, automated quality inspection has become crucial for improving production efficiency. Consequently, accurate measurement of fabric defect sizes remains highly valuable in current research. Although the proposed method demonstrates good performance in size estimation, challenges persist in detecting defects with weak features. Moreover, defect measurement for printed fabrics has yet to be widely explored, which remains a pressing issue in industrial practice. Therefore, future work will further extend the proposed measurement framework by improving recall and accuracy for subtle defects in plain fabrics, as well as expanding the approach to include defect size estimation in printed fabrics. This aims to promote the development of efficient and robust measurement methods under complex textured backgrounds, better aligning with the practical needs of accurate fabric quality assessment.

References

[1] V. Azamfirei, F. Psarommatis, and Y. Lagrosen, "Application of automation for in-line quality inspection, a zero-defect manufacturing approach," *J. Manuf. Syst.*, vol. 67, pp. 1–22, Apr. 2023.

[2] S. Zhao, R. Y. Zhong, J. Wang, C. Xu, and J. Zhang, "Unsupervised fabric defects detection based on spatial domain saliency and features clustering," *Comput. Ind. Eng.*, vol. 185, Nov. 2023, Art. no. 109681.

[3] Y. Huang, J. Jing, and Z. Wang, "Fabric defect segmentation method based on deep learning," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–15, 2021.

[4] J. Wang, J. Cheng, C. Gao, J. Zhou, and L. Shen, "Enhanced fabric defect detection with feature contrast interference suppression," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025.

[5] L. Shao, E. Zhang, Q. Ma, and M. Li, "Pixel-wise semisupervised fabric defect detection method combined with multitask mean teacher," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.

[6] B. Fang, X. Long, F. Sun, H. Liu, S. Zhang, and C. Fang, "Tactile-based fabric defect detection using convolutional neural network with attention mechanism," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–9, 2022.

[7] R. Chen et al., "Patch matching for few-shot industrial defect detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–11, 2024.

[8] S. Su, S. Du, D. Wang, and X. Lu, "Toward better generalization: Shape feature-enhanced fastener defect detection with diffusion model," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–13, 2024.

[9] Y. Jin, Y. Lu, G. Zhou, Q. Liu, and Y. Wang, "Glass wool defect detection using an improved YOLOv5," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 4385–4394.

[10] H. Yang et al., "CSRM-MIM: A self-supervised pretraining method for detecting catenary support components in electrified railways," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 4, pp. 10025–10037, Aug. 2025.

[11] J. Yan et al., "Research on multimodal techniques for arc detection in railway systems with limited data," *Struct. Health Monitor.*, early access, May 2025, Art. no. 14759217251336797.

[12] Z. Ren, F. Fang, N. Yan, and Y. Wu, "State of the art in defect detection based on machine vision," *Int. J. Precis. Eng. Manuf.-Green Technol.*, vol. 9, no. 2, pp. 661–691, Mar. 2022.

[13] V. D. Truong, J. Xia, Y. Jeong, and J. Yoon, "An automatic machine vision-based algorithm for inspection of hardwood flooring defects during manufacturing," *Eng. Appl. Artif. Intell.*, vol. 123, Aug. 2023, Art. no. 106268.

[14] Z. Yin, L. Zhao, J. Cheng, H. Liu, and M. Chen, "Rapid inspection system for weak feature micro-flaw on large-aperture optics surface based on multi illumination modes and multitask learning algorithms," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.

[15] Z. Yin et al., "A novel automatic classification approach for micro-flaws on the large-aperture optics surface based on multi-light source fusion and integrated deep learning architecture," *J. Intell. Manuf.*, vol. 35, no. 1, pp. 413–428, Jan. 2024.

[16] F. Li, K. Xiao, Z. Hu, and G. Zhang, "Fabric defect detection algorithm based on improved YOLOv5," *Vis. Comput.*, vol. 40, no. 4, pp. 2309–2324, Apr. 2024.

[17] S. Zhao, R. Y. Zhong, C. Xu, J. Wang, and J. Zhang, "A dynamic inference network (DI-Net) for online fabric defect detection in smart manufacturing," *J. Intell. Manuf.*, vol. 36, no. 4, pp. 1–16, Apr. 2025.

[18] C. Zhang, S. Feng, X. Wang, and Y. Wang, "ZJU-leaper: A benchmark dataset for fabric defect detection and a comparative study," *IEEE Trans. Artif. Intell.*, vol. 1, no. 3, pp. 219–232, Dec. 2020.

[19] M. S. Hesarian, M. Eshkevari, and M. J. Rezaee, "Angle analysis of fabric wrinkle by projected profile light line method, image processing and neuro-fuzzy system," *Int. J. Comput. Integr. Manuf.*, vol. 33, nos. 10–11, pp. 1167–1184, Nov. 2020.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.

[20] M. Chen et al., "Improved faster R-CNN for fabric defect detection based on Gabor filter with genetic algorithm optimization," *Comput. Ind.*, vol. 134, Jan. 2022, Art. no. 103551.
[21] I. Koulali and M. T. Eskil, "Unsupervised textile defect detection using convolutional neural networks," *Appl. Soft Comput.*, vol. 113, Dec. 2021, Art. no. 107913.
[22] S. Shakir and C. Topal, "Unsupervised fabric defect detection with local spectra refinement (LSR)," *Neural Comput. Appl.*, vol. 36, no. 3, pp. 1091–1103, Jan. 2024.
[23] J. Li and X. Kang, "Mobile-YOLO: An accurate and efficient three-stage cascaded network for online fiberglass fabric defect detection," *Eng. Appl. Artif. Intell.*, vol. 134, Aug. 2024, Art. no. 108690.
[24] Y. Kai, L. Wentao, Y. Xuyi, G. Qing, X. Weiqiang, and L. Zhang, "FA-YOLO: A high-precision and efficient method for fabric defect detection in textile industry," *IEICE Trans. Fundam. Electron., Commun. Comput. Sci.*, vol. E107.A, no. 6, pp. 890–898, 2024.
[25] H. Qu, L. Di, J. Liang, and H. Liu, "U-SMR: U-SwinT & multi-residual network for fabric defect detection," *Eng. Appl. Artif. Intell.*, vol. 126, Nov. 2023, Art. no. 107094.
[26] T. Defard, A. Setkov, A. Loesch, and R. Audigier, "PaDiM: A patch distribution modeling framework for anomaly detection and localization," in *Proc. Int. Conf. Pattern Recognit.*Cham, Switzerland: Springer, Jan. 2021, pp. 475–489.
[27] S. Lee, S. Lee, and B. C. Song, "CFA: Coupled-hypersphere-based feature adaptation for target-oriented anomaly localization," *IEEE Access*, vol. 10, pp. 78446–78454, 2022.
[28] W. Luo et al., "Exploring intrinsic normal prototypes within a single image for universal anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 9974–9983.
[29] J. Guo, S. Lu, W. Zhang, F. Chen, H. Li, and H. Liao, "Dinomaly: The less is more philosophy in multi-class unsupervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 20405–20415.
[30] W. Liu et al., "Towards visually explaining variational autoencoders," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8642–8651.
[31] Y. Liang, J. Zhang, S. Zhao, R. Wu, Y. Liu, and S. Pan, "Omni-frequency channel-selection representations for unsupervised anomaly detection," *IEEE Trans. Image Process.*, vol. 32, pp. 4327–4340, 2023.
[32] A. Mousakhan, T. Brox, and J. Tayyub, "Anomaly detection with conditioned denoising diffusion models," in *Proc. DAGM German Conf. Pattern Recognit.*, 2023, pp. 181–195.
[33] C.-L. Li, K. Sohn, J. Yoon, and T. Pfister, "CutPaste: Self-supervised learning for anomaly detection and localization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9664–9674.
[34] H. M. Schlüter, J. Tan, B. Hou, and B. Kainz, "Natural synthetic anomalies for self-supervised anomaly detection and localization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 474–489.
[35] C. Gao, X. Chen, J. Zhou, J. Wang, and L. Shen, "Open-set fabric defect detection with defect generation and transfer," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–13, 2025.
[36] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, "Towards total recall in industrial anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14298–14308.
[37] Z. Gu et al., "Remembering normality: Memory-guided knowledge distillation for unsupervised anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16401–16409.
[38] J. Lei, X. Hu, Y. Wang, and D. Liu, "PyramidFlow: High-resolution defect contrastive localization using pyramid normalizing flow," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14143–14152.
[39] J. Yu et al., "FastFlow: Unsupervised anomaly detection and localization via 2D normalizing flows," 2021, *arXiv:2111.07677*.
[40] S. Wei, J. Jiang, and X. Xu, "UniNet: A contrastive learning-guided unified framework with feature selection for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 9994–10003.
[41] K. Batzner, L. Heckler, and R. König, "EfficientAD: Accurate visual anomaly detection at millisecond-level latencies," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 128–138.
[42] M. Rudolph, T. Wehrbein, B. Rosenhahn, and B. Wandt, "Asymmetric student-teacher networks for industrial anomaly detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 2591–2601.
[43] G. Wang, S. Han, E. Ding, and D. Huang, "Student–teacher feature pyramid matching for anomaly detection," 2021, *arXiv:2103.04257*.
[44] T. D. Tien et al., "Revisiting reverse distillation for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 24511–24520.
[45] H. Deng and X. Li, "Anomaly detection via reverse distillation from one-class embedding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9737–9746.
[46] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12993–13000.
[47] H. Fu, W. Huang, Z. Li, Y. Cheng, and T. Zhang, "Robust cubature Kalman filter with Gaussian-multivariate Laplacian mixture distribution and partial variational Bayesian method," *IEEE Trans. Signal Process.*, vol. 71, pp. 847–858, 2023.
[48] A.-T. Ardelean and T. Weyrich, "High-fidelity zero-shot texture anomaly localization using feature correspondence analysis," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 1134–1144.

Shangyan Zhang is currently pursuing the Ph.D. degree with Jiangnan University, Wuxi, China.

His main research interests include textile image processing and fabric defect size measurement.

Jian Zhou received the Ph.D. degree in textile science and engineering from Donghua University, Shanghai, China, in 2014.

He is currently an Associate Research Professor with the College of Textile Science and Engineering, Jiangnan University, Wuxi, China. His main research interests include intelligent detection of textiles, fabric defect detection, and intelligent manufacturing of textiles.

Jun Xiang received the Ph.D. degree in textile science and engineering from Jiangnan University, Wuxi, China, in 2022.

He is currently an Associate Research Professor with the College of Textile Science and Engineering, Jiangnan University. His main research interests include intelligent detection of textiles, textile graphic simulation, and intelligent manufacturing of textiles.

Ruru Pan received the B.S. and Ph.D. degrees in textile engineering from Jiangnan University, Wuxi, China, in 2005 and 2010, respectively.

He is currently a Professor with the College of Textile Science and Engineering, Jiangnan University. His main research interests include digital textile technology and digital image processing of textile.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:59:31 UTC from IEEE Xplore. Restrictions apply.